Third-cumulant lift assessment for the SC-SCOPE hypothesis: the cubic sunset channel is anchor-subdominant but endpoint-critical under conservative bounds

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

The U1 candidate theorem carries SC-SCOPE (matched second-cumulant bookkeeping) as a named hypothesis; unit U3 identified its lift as the RES-5 frontier. This note assesses WHAT the third-cumulant order adds: which new channels appear, their dressed couplings, and their size at the three anchor intensities relative to the U1 composed margins. The output is an honest feasibility map for the lift, NOT a lift: every bound below is a conservative sup-kernel ESTIMATE on certified constants, pending machine verification.

2. The new channels at third order

At second cumulant the fluctuation correction is governed by $W = \lambda'(F^2 - 2I)$ (multiplication; the P^2 theorem). Expanding the interaction about the condensate F to the next cumulant order, the fluctuation Hamiltonian acquires the CUBIC vertex

$$V_3 \;=\; \int g_3(x)\,\delta\varphi^3(x), \qquad g_3(x) \;=\; u_{\mathrm{eff}}(M)\,F(x) \;+\; \tfrac{10v}{3}F(x)^3,$$

where the dressing $u \to u_{\mathrm{eff}}(M) = u + 10vM$ arises from the Gaussian contraction of the $F\,\delta\varphi^5$ sextic term ($vF\delta\varphi^5 \to 10vG(0)\,F\delta\varphi^3$, $G(0) = M$) — exactly the same dressing as the condensate quartic. At the anchor dressing $M = M_R = 0.109414$:

$$u_{\mathrm{eff}}(M_R) \;=\; -0.86 + 32.4 \times 0.109414 \;=\; +2.6850, \qquad u_{\mathrm{eff}}^2 = 7.209,$$

a factor 9.7 ABOVE $u^2 = 0.7396$: the bare-coupling estimate is misleadingly optimistic and is not used below.

The third-cumulant free-energy contribution of V_3 is

$$\delta F_3 \;=\; -\tfrac{1}{2}\,\langle V_3^2\rangle_c \;=\; -\tfrac{1}{2}\sum_t |\hat g_3(t)|^2\left[\,6\,\widehat{G^3}(t) + 9\,M^2\,\hat G(t)\,\right],$$

with $\widehat{G^3} = G{*}G{*}G$ (the sunset kernel) and the $9M^2\hat G$ piece the tadpole-dressed channel. Two structural facts:

- **One-sidedness**: $\delta F_3 \le 0$ always (second-order perturbation lowers the free energy) and $\delta F_3[\mathcal{R}_H] = 0$ (no condensate $\Rightarrow g_3 \equiv 0$): the cubic channel can ONLY help competitors. It is therefore the exactly relevant threat to the selection margin.

- **Tadpole status**: at the variational stationary point the $9M^2\hat{G}$ piece pairs with the condensate-equation counterterm (the $3G(0)g_3$ effective linear source that the Hartree-dressed gap/ condensate equations absorb); whether this cancellation survives the MATCHED bookkeeping for non-stationary competitor evaluations is a statement that must be PROVEN for the lift — it is load-bearing (see section 4).

3. Conservative size bounds (ESTIMATE grade)

Kernel sup-bounds on certified constants: $\widehat{G^3}(t) = \int (G{*}G)(k)\,G(t-k) \le \sup_s (G{*}G)(s)\cdot\int G = J(0)\,M_R$ with $J(0) = 0.290$ (monotone envelope, AddC) and $\int G = M(\hat{r}) \le M_R$ (safe direction: $\hat{r} > m^*$); $\hat{G}(t) \le G_{\max} = 1/\hat{r} = 3.215$. Parseval: $\sum_t |\hat{g}_3(t)|^2 = \langle g_3^2\rangle \approx u_{\text{eff}}^2\,\langle F^2\rangle = u_{\text{eff}}^2\,2I$ (the F^3 and cross terms contribute $\le 15\%$ at the endpoint, $\le 3\%$ at the anchor; folded as headroom, see sanity check).

Quantity	$I = 4{\times}10^{-4}$	$I = 10^{-3}$	$I = 2{\times}10^{-3}$
sunset bound $\frac{1}{2}u_{\text{eff}}^2 2I \cdot 6J(0)M_R$	5.49×10^{-4}	1.37×10^{-3}	2.75×10^{-3}
composed U1 margin $0.00432(1-1/\rho)$	4.25×10^{-3}	3.83×10^{-3}	2.66×10^{-3}
ratio (sunset)	$\times 7.7$	$\times 2.8$	$\times 0.97$ (FAILS)
tadpole bound $\frac{1}{2}u_{\text{eff}}^2 2I \cdot 9M_R^2 G_{\max}$	9.99×10^{-4}	2.50×10^{-3}	5.00×10^{-3}
ratio (tadpole, if uncancelled)	$\times 4.3$	$\times 1.5$	$\times 0.53$ (FAILS)

($\rho =$ the hardened/recorded STEP-5B margin ratios 59.4/8.8/2.6; the middle uses the AddD figure since AddE quoted the two band edges.)

4. Honest verdict and the lift requirements

- **Production anchor**: the third-cumulant cubic channel is margin-subdominant under conservative sup-bounds ($\times 7.7$ sunset; $\times 4.3$ tadpole-if-uncancelled). The SC-SCOPE lift at the anchor looks FEASIBLE by the present route.

- **Endpoint** $I = 2{\times}10^{-3}$: the conservative bounds FAIL ($\times 0.97$ / $\times 0.53$). This does NOT mean the selection fails there — it means sup-kernel estimates are too crude at the endpoint, and the lift REQUIRES the chain's sharper machinery: (a) per-transfer decay $\widehat{G^3}(|t|)$ evaluated on the realized cubic transfers (which obey $|t| \ge q_0$-scale separations on the amended class, where $\widehat{G^3}$ is far below its sup — the same refinement that AddE's J_{eff} delivered at second order); (b) cubic-transfer counting (the analogue of the $K(n)$ energy bound: $\hat{g}_3$ is supported on $O(n)$ first-shell modes plus $O(n^2)$ F^3-fibers with $I^{3/2}$-suppressed weights, so the Parseval mass overcounts the J-weighted sum); (c) the TADPOLE CANCELLATION statement at matched bookkeeping — load-bearing: without it the lift fails at $I \ge 10^{-3}$ within $\times 1.5$.

- **One more channel flagged**: the quartic-vertex second-order channel ($\langle(\delta\varphi^4)^2\rangle$-type, present for BOTH $\mathcal{R}_H$ and competitors) contributes to the DIFFERENCE only through its F-dependent dressing; its matched-order cancellation structure must be written out in the lift (registered as part of the same follow-up).

The assessment sharpens SC-SCOPE from a formal bookkeeping fence into a quantified hypothesis: its lift is a real (not cosmetic) programme whose anchor-feasibility is supported and whose endpoint requires exactly three named technical inputs.

5. Numerical verification

ALL numbers in this note are ESTIMATE-grade closed-form arithmetic on certified constants ($u = -0.86$, $v = 3.24$, $M_R = 0.109414$, $J(0) = 0.290$, $\hat{r} = 0.3110$, margins 59.4/8.8/2.6, band floor 0.00432) from the existing artefact `claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/result.json` and the AddC/AddE notes. NO machine asserts have been executed for the new combinations (sandbox shell unavailable this session); the follow-up script is registered (planned: codes/vacuum/third_cumulant_assessment.py, with self-test asserts for every table entry and a JSON artefact under claims/B5-BEYOND-LAYER-BOUND/runs/).

Quantitative sanity check (mandatory): dimensional — $\delta F_3 \sim (\text{coupling})^2 \times (\text{field}^2) \times (\text{kernel})$ with the kernel $J(0)M_R$ carrying the two-loop measure, matching the free-energy-density units of the margin; magnitude — the headroom check: the neglected $\hat{g}_3$ pieces are the cross term $2|u_{\text{eff}}| \cdot \frac{10v}{3}\langle F^4\rangle \le 2 \cdot 2.685 \cdot 10.8 \cdot 12I^2$ and the F^3 square $\le 116.6 \cdot \langle F^6\rangle$ with $\langle F^6\rangle = O(I^3)$; at $I = 2 \times 10^{-3}$ these total $\le 15\%$ of the leading $u_{\text{eff}}^2\, 2I$ term — the quoted endpoint ratio 0.97 would tighten to ~ 0.84 with them included, NOT loosen (the failure finding is robust in the safe direction); limit case — as $I \to 0$ every third-cumulant bound vanishes linearly in I while the margin is fixed, recovering the trivial low-intensity limit; sign-direction — $\delta F_3 \le 0$ for competitors and $= 0$ for $\mathcal{R}_H$ is the dangerous direction and is treated as such (the bound is applied AGAINST the margin, never in its favor).

6. Devil's-advocate (three objections; CLAUDE.md 6.3)

(α) *"The dressed coupling $u_{\text{eff}}(M_R)$ might itself be wrong for the cubic vertex: the sextic contraction count $10vG(0)$ deserves a derivation, not an analogy to the quartic dressing."* VALID-with-mitigation: the count is $\binom{5}{2}\cdot$(pairings) $vF\delta\varphi^5 \to 10vM\,F\delta\varphi^3$, the same Wick combinatorics as the certified condensate quartic dressing ($u_{\text{eff}} = u + 10vM$ appears verbatim in Math437 Lemma 1); the follow-up script will verify the coefficient by symbolic Wick enumeration. Until then the note is T2 ESTIMATE and says so.

(β) *"The endpoint FAILURE of the conservative bound could be an artifact of double conservatism: sup-kernel AND full Parseval mass AND uncancelled tadpole stacked together."* DISMISSED as an objection to the note's claim — because the note CLAIMS exactly this: the finding is 'conservative bounds fail at the endpoint, hence the lift requires the sharper machinery (a)/(b)/(c)', not 'the selection fails at the endpoint'. The stacking is deliberate: it identifies which refinements are load-bearing. The one-sided headroom check (sanity section) confirms the failure direction is robust to the neglected terms.

(γ) *"Comparing δF_3 against the COMPOSED U1 margin double-penalizes: the second-cumulant beyond-layer budget already consumed $1/\rho$ of the floor; maybe the right comparison is against the full 0.00432."* VALID-with-mitigation: both comparisons are reported by construction (against the full floor the endpoint sunset ratio is $0.00432/2.75 \times 10^{-3} = \times 1.57$ — still thin); the composed-margin comparison is the correct one for the U1 theorem as stated (the budgets must COEXIST), while the full-floor comparison is the right one if the lift re-derives a joint second+third-order budget in one inequality. The follow-up lift should do the latter; the assessment flags both.

(δ) *"$\int G = M(\hat{r})$ was bounded by $M_R = M(m^*)$ using $\hat{r} > m^*$; verify the direction."* DISMISSED: $\hat{r} = 0.3110 > m^* = 0.3045$ and M is strictly decreasing (unit U2, L1), so $M(\hat{r}) < M(m^*) = M_R$ — the substitution ENLARGES the bound (safe direction).

7. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / third-cumulant-lift-assessment (unit U4)
Precise statement:	ESTIMATE (T2): the new third-cumulant channel is the cubic sunset with DRESSED coupling u_eff(M_R) = +2.685 (x9.7 above u^2 in square); conservative sup-bounds give sunset/margin ratios x7.7 / x2.8 / x0.97 and tadpole-if-uncancelled x4.3 / x1.5 / x0.53 at I = 4e-4 / 1e-3 / 2e-3: anchor-feasible, ENDPOINT-CRITICAL. Lift requirements named: (a) per-transfer G^3-kernel decay, (b) cubic-transfer counting, (c) tadpole cancellation at matched bookkeeping (load-bearing).
Scope:	SC-SCOPE quantification only; no lift is performed; no change to any theorem, tier, or margin. Amended class, three anchors.
Dependencies:	U1 composed margins; AddC (J(0)), AddE (hardened ratios); Math437 closed forms (u_eff dressing); U2 L1 (monotonicity, for one bound direction); certified anchor constants.
Evidence grade:	ESTIMATE (T2) -- closed-form arithmetic on certified constants; machine asserts NOT yet executed (follow-up script registered).
Reproduction command:	(registered follow-up) python codes/vacuum/third_cumulant_assessment.py
Expected output:	asserts for every table entry (u_eff, kernel sups, six ratios) + JSON artefact under claims/B5-BEYOND-LAYER-BOUND/runs/.
Falsification gate:	Machine verification contradicting any table entry by > 5% (would demote the assessment); a proof that the tadpole channel does NOT cancel at matched bookkeeping AND a sharp per-transfer bound still exceeding the endpoint margin (would mean SC-SCOPE cannot be lifted at the endpoint by this route -- honest negative pre-registered).
Tier before / after:	No tier action. SC-SCOPE remains a named hypothesis of the U1 candidate set; this note quantifies it.
No-overclaim statement:	Nothing here lifts SC-SCOPE or threatens the U1 conditional theorem (which is AT second cumulant by hypothesis). The endpoint bound failure is a statement about the crudeness of sup-estimates, pre-registered honestly, NOT a counterexample. All numbers pending machine

verification. PDF build + linter PENDING operator-side (sandbox shell unavailable).

Next required action:	Follow-up script (machine asserts); then the three named lift inputs (a)/(b)/(c) as separate units if the operator directs the SC-SCOPE lift; mainline continues with U5 (B5 full-T5 dossier).